SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3 ASSESSMENT TOOL 1 – Git & Docker Technical Assignment

Course Code:	CSA10 / CSA1007	**Course Title:**	Software Engineering
CO Assessed:	CO3	**Assessment:**	Assessment Tool 1 – Git & Docker Technical Assignment
Weightage:	20%	**Total Marks:**	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
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Date:	11/08/2026	**Duration:**	60 minutes

Code of Conduct

I Sk. Parvez Ahamed (192411084) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sk. Parvez Ahamed__

Problem Statement 13: Blockchain-Based Academic Credential Verification Platform

1.Abstract

The Blockchain-Based Academic Credential Verification Platform is a web-based software system designed to help academic institutions issue, store, and verify degree certificates, mark sheets, and other credentials in a tamper-proof and independently verifiable manner. The platform records a cryptographic proof of every issued credential on a blockchain network, so that employers, universities, and other verifiers can confirm the authenticity of a credential instantly, without depending on manual attestation, phone verification, or physical document exchange. The system provides a centralized dashboard for institutions to issue credentials, monitor verification requests, and manage revocations, while the underlying smart contract guarantees that once a credential is recorded, it cannot be silently altered. Git and GitHub are used for version control, collaborative development, feature branching, meaningful commits, merging, and remote repository synchronization. Docker is used to package the smart contract environment, backend service, and database into a consistent runtime environment, and Docker Compose is used to define the application services and make local deployment repeatable.

2.Problem Statement

Academic credentials such as degree certificates, transcripts, and course completion records are frequently forged, altered, or misrepresented, which creates serious challenges for employers, universities, and government agencies during background verification. Traditional verification methods rely on manual cross-checking with the issuing institution, physical seals and signatures, or slow email-based confirmation, all of which are time-consuming, inconsistent, and vulnerable to fraud. When credential records are maintained only in institutional databases or paper form, there is no independent, tamper-evident way for a third party to confirm that a certificate is genuine and has not been altered after issuance. The proposed Blockchain-Based Academic Credential Verification Platform addresses this gap by anchoring a verifiable digital fingerprint of every credential on a blockchain, giving institutions a fast issuance workflow and giving verifiers an instant, trustworthy way to check authenticity and revocation status.

3.Objectives

- ❖ Develop a clear web dashboard for institutions to issue and manage academic credentials.
- ❖ Maintain structured credential records including student identity, institution, course/degree, and issue date.
- ❖ Record a verifiable hash of every credential on a blockchain network and expose its verification status.
- ❖ Demonstrate Git branching, meaningful commits, merging, and GitHub synchronization.
- ❖ Containerize the application, blockchain node, and database using Docker for consistent execution.
- ❖ Use Docker Compose for repeatable local deployment and multi-service configuration.
- ❖ Test the application through a browser and verify smart contract execution on the local chain.

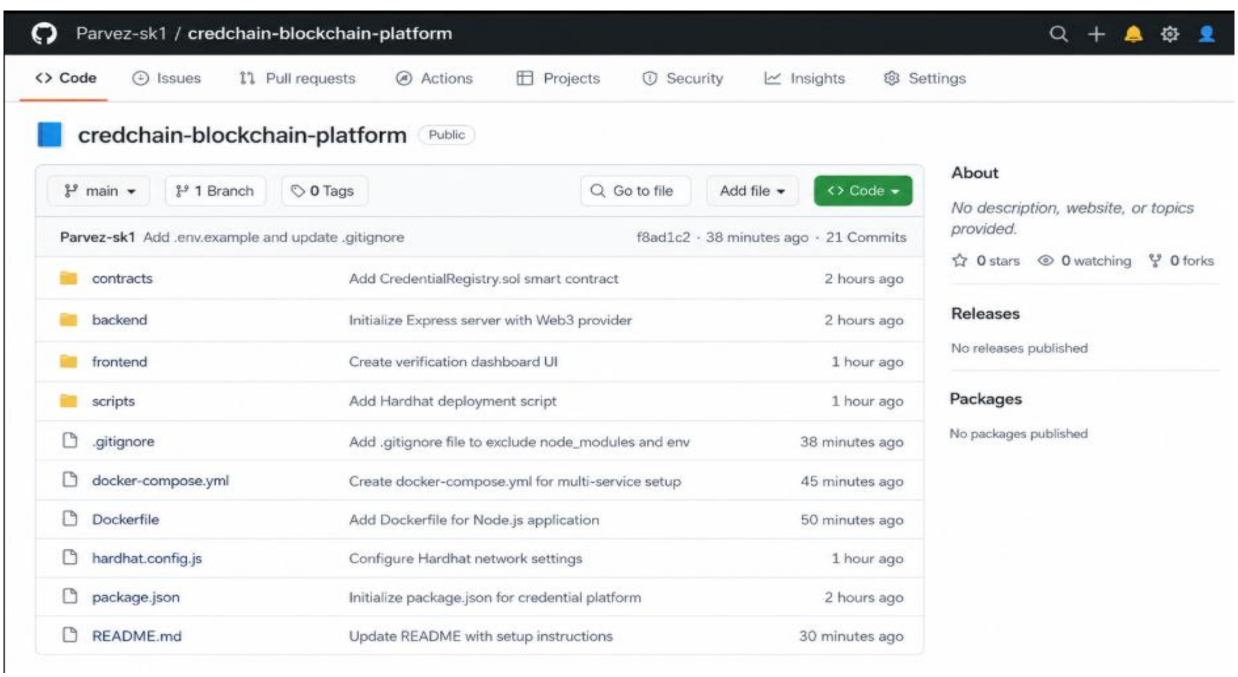

Fig 1: GitHub

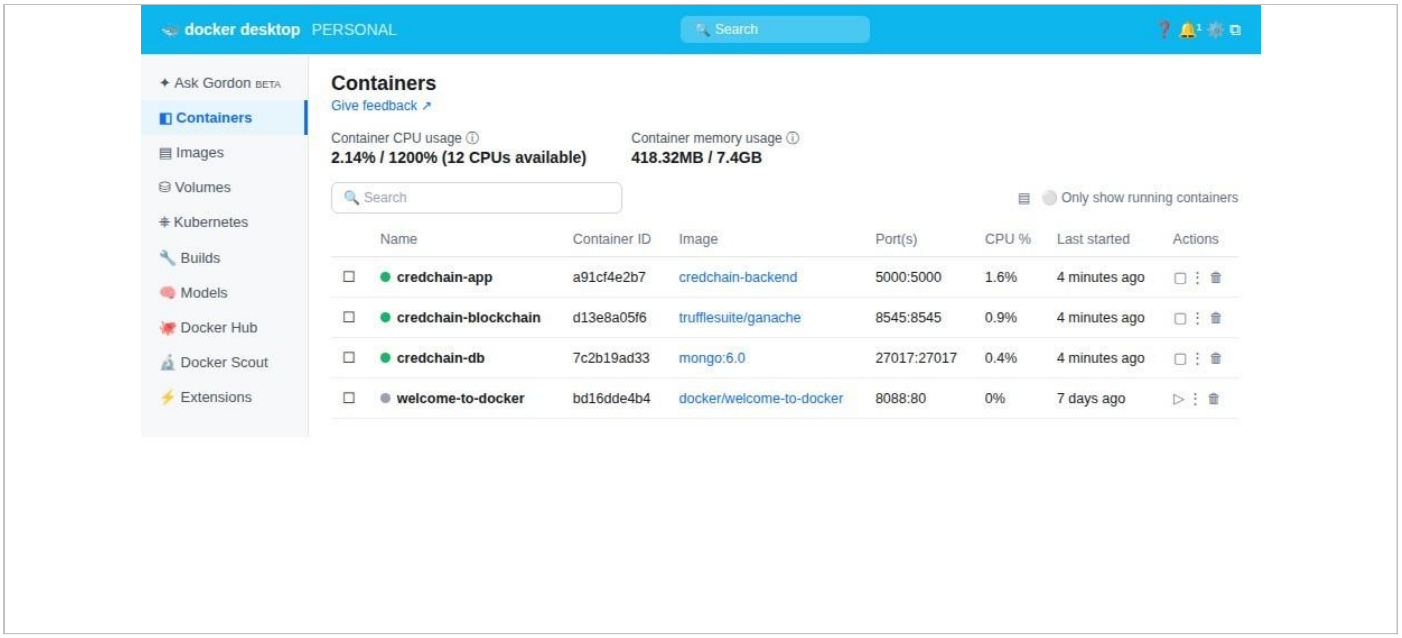

Fig 2: Docker Implementation

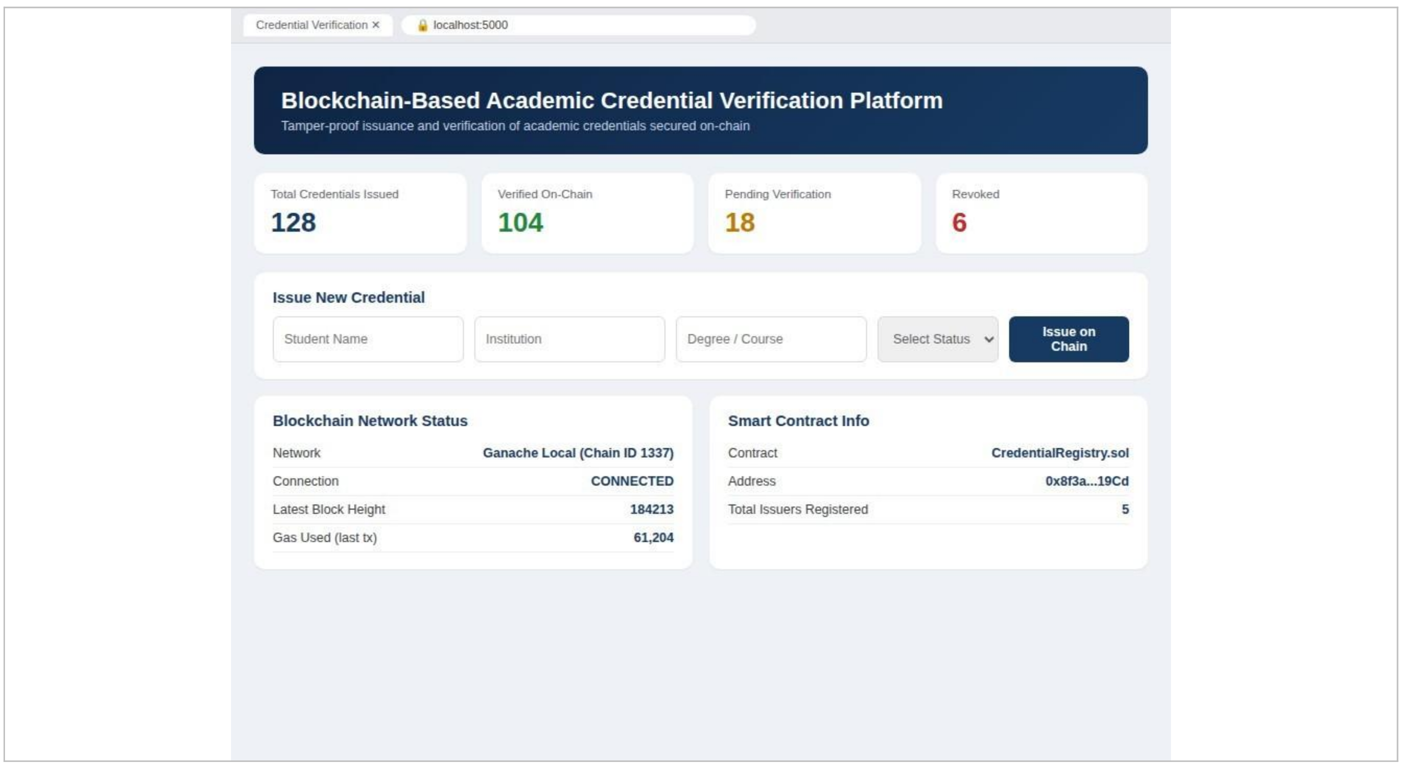

Fig 3a: Dashboard

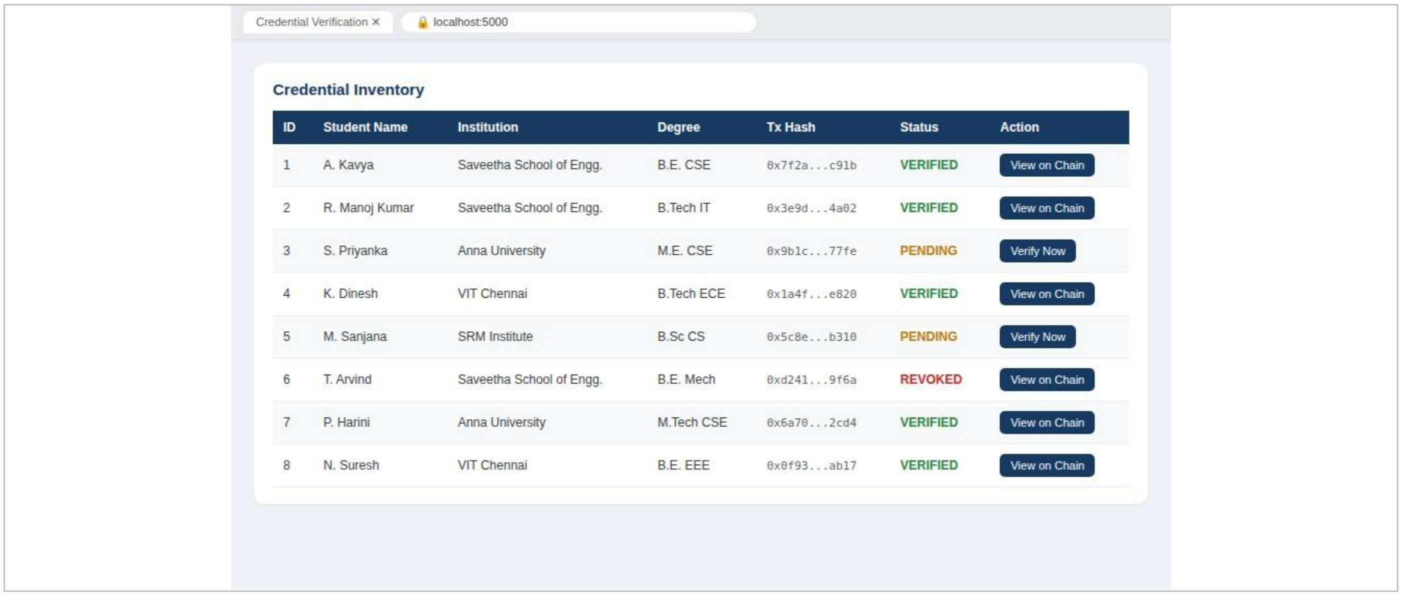

Fig 3b: Dashboard

4.Technologies Used

Technology / Tool	Purpose	Details
Solidity	Smart contract logic for credential issuance, verification, and revocation	^0.8.x
Web3.py	Connects Python backend to blockchain	Python blockchain library
Python / Flask	Backend API server and application logic	Python 3.x, Flask
MongoDB	Off-chain storage of student and credential records	MongoDB 6.0
HTML / CSS / JavaScript	Frontend interface and verification dashboard	Web standards
Git / GitHub	Version control and remote repository	Remote Git hosting
Docker / Docker Compose	Containerization and multi-service deployment	Docker Desktop, Compose v2
VS Code	Development and terminal environment	Desktop IDE

5.Application Workflow

1. The project is developed and maintained using VS Code.
2. Git tracks the smart contract source, backend application, configuration, and documentation.

3. The Express backend receives browser requests and serves the credential verification dashboard.
4. New credential data entered by the institution is hashed and submitted to the CredentialRegistry smart contract.
5. The smart contract stores the credential hash, issuer address, and timestamp immutably on the local blockchain.
6. Off-chain metadata (student name, institution, course) is stored in MongoDB and linked to the on-chain transaction hash.
7. The dashboard displays credential status (Verified, Pending, Revoked) by querying the blockchain and the database.
8. Docker builds an image containing the backend application and its dependencies.
9. Docker Compose starts the application, blockchain node, and database services together and maps the application port.
10. The user opens the local application in a web browser and verifies the dashboard and on-chain transaction details.

6.Container Deployment Commands

- ➢ docker build -t credchain-app .
- ➢ docker run -p 5000:5000 credchain-app
- ➢ docker compose up --build
- ➢ docker ps
- ➢ docker logs credchain-app
- ➢ docker compose down

7.Deployment Workflow

1. Build the Docker image from the project directory.
2. Start all services (application, blockchain node, database) using Docker Compose.
3. Check the running containers using docker ps.
4. Open the application using the browser at http://localhost:5000/.
5. Verify dashboard pages, credential records, verification status, and blockchain transaction hashes.
6. Inspect container logs if any application, database, or blockchain connection issue occurs.
7. Stop the deployment using docker compose down after testing.

8.Conclusion

The Blockchain-Based Academic Credential Verification Platform demonstrates the complete lifecycle of a small software project: problem analysis, system design, application development, version control, feature branching, merging, containerization, deployment, and testing. The platform provides a centralized and tamper-evident approach to credential issuance, verification, and revocation, while removing the need for slow manual attestation between institutions and verifiers. Git and GitHub support organized development and traceable project history, while Docker and Docker Compose provide a consistent and repeatable deployment environment for the backend service, blockchain node, and database. The proposed architecture also provides a strong foundation for future enhancements such as decentralized storage of certificate documents (e.g., IPFS), multi-institution issuer onboarding, mobile-based QR verification, and integration with public blockchain networks for wider trust and adoption.